The International Credit Channel of U.S. Financial Shocks

Discussion of Cesa-Bianchi & Sokol (2022)

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Advanced Macro
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Introduction

Is there an international credit channel beyond monetary policy?

- U.S. monetary policy transmits across borders: synchronised movements in credit spreads, capital flows, and asset prices – the **International Credit Channel** (Rey, 2016).
- U.S. shocks affect net worth of agents worldwide ⇒ co-movement of financial conditions.
- **This paper:** does the same credit channel transmit *financial* shocks internationally – and how does it compare to monetary policy (MP) and central-bank information (CBI) shocks?

Financial shocks explain business cycles

- **Financial shocks**, broadly defined: exogenous changes to the net worth of financially constrained agents, or to their degree of risk aversion.
- **Identification is hard** – they mimic
  - central-bank information shocks, and
  - aggregate-demand shocks.
- Need an approach that **separately identifies** financial shocks to assess their domestic and international effects.

Novel identification: combining IVs and sign restrictions

1. **Step 1.** Jointly identify MP and CBI shocks via high-frequency FOMC surprises (Jarocinski & Karadi, 2020) and use them as **external instruments**.
2. **Step 2.** Conditional on Step 1, derive a **minimum set of sign restrictions** from theory that uniquely identify a broad class of financial shocks.

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$\Rightarrow$ Embedded in a **two-country SVAR** for the U.S. and U.K. *(large small-open economy, floating ER, inflation targeting, well-developed financial sector).*

Main results

1. **Existence of an ICC in U.S. financial-shock transmission** – U.S. shocks tighten domestic credit, transmit to U.K. spreads, and lower U.K. real activity.

   Extends Helbling et al. (2011), Eickmeier & Ng (2015) – we pin down the credit-channel mechanism.

2. **U.S. MP also transmits internationally via credit spreads** – narrowing the U.S.–U.K. spread differential.

   Complements Rey (2016), Passari & Rey, Gerko & Rey (2017), Kalemli-Özcan (2019) – we control for CBI.

3. **CBI shocks generate cross-country spillovers via the ICC** – a positive U.S. outlook compresses U.S. spreads and transmits quickly to the U.K.

   Complements Stavrakeva & Tang (2019), Miranda-Agrippino & Rey (2020), Jarocinski (2020).

Presentation Roadmap

1. **Introduction**
2. **Econometric Approach**
   - Identification strategy: IVs + sign restrictions
   - Two-country SVAR
   - Identifying financial shocks
3. **Data & Results**
   - Domestic and international transmission
4. **Mechanism**
5. **Conclusion**

Econometric Approach

Identification strategy

Standard SVAR:

$$y_t = \Phi(L)\, y_{t-1} + B\, e_t, \qquad \mathbb{E}[e_t e_t'] = I_n$$

- y_t: $n \times 1$ vector of observables; e_t: unobserved structural shocks; B: impact matrix; $u_t = Be_t$.
- Identification problem: $\Sigma_u = BB'$ gives $n(n+1)/2$ equations but n^2 unknowns in B.

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Partition into two types of shocks:

$$B = \begin{bmatrix} B^{IV} & B^{SR} \end{bmatrix}, \qquad e_t^{IV} = (e_{1,t}', \ldots, e_{k,t}')', \quad e_t^{SR} = (e_{k+1,t}', \ldots, e_{n,t}')'$$

- $k < n$ shocks identified by **external instruments** – B^{IV} is $n \times k$.
- Remaining $n - k$ shocks identified by **sign restrictions** – B^{SR} is $n \times (n-k)$.
- Here $k = 2$: one monetary-policy shock and one CB-information shock.

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External instruments: $z_t = (z_{1,t}', \ldots, z_{k,t}')'$

$$\mathbb{E}[z_t e_t^{IV\prime}] \neq 0, \qquad \mathbb{E}[z_t e_t^{SR\prime}] = 0$$

MP surprise + CBI surprise from high-frequency Fed-Funds futures around FOMC announcements (Jarocinski & Karadi, 2020).

Combining sign restrictions and external instruments

Rely on economic theory to pin down a **unique** set of sign restrictions consistent with e_t^{SR}.

$$\Sigma_u = BB' = \begin{bmatrix} B^{IV} & B^{SR} \end{bmatrix} \begin{bmatrix} B^{IV} & B^{SR} \end{bmatrix}'$$

$$\Sigma_u = CC' = CQQ'C' = (CQ)(CQ)', \quad \text{where } QQ' = I,\ C \text{ is Cholesky of } \Sigma_u$$

$$CQ = \begin{bmatrix} B^{IV} & B^{SR} \end{bmatrix} = B$$

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Algorithm (Rubio-Ramirez et al., 2010):

1. Find q such that $Cq = B^{IV}$ – IV pins down the first k columns exactly.
2. Complete Q via Gram-Schmidt over the remaining $n - k$ columns.
3. Retain the draw if B^{SR} satisfies sign restrictions; discard otherwise.

Repeat over bootstrap draws – report 90% credible intervals over all retained Q.

A two-country SVAR

Setup: U.S. = Foreign (F), U.K. = Home (H):

$$\begin{bmatrix} y_t^F \\ y_t^H \end{bmatrix} = \begin{bmatrix} \Phi_{FF}(L) & \Phi_{FH}(L) \\ \Phi_{HF}(L) & \Phi_{HH}(L) \end{bmatrix} \begin{bmatrix} y_{t-1}^F \\ y_{t-1}^H \end{bmatrix} + \begin{bmatrix} B_{FF} & B_{FH} \\ B_{HF} & B_{HH} \end{bmatrix} \begin{bmatrix} e_t^F \\ e_t^H \end{bmatrix}$$

where $y_t = (y_t^{F\prime}, y_t^{H\prime})'$ collects $n = n_F + n_H$ endogenous variables, $e_t = (e_t^{F\prime}, e_t^{H\prime})'$ are structural shocks with $\mathbb{E}[e_t e_t'] = I$, and $\Phi(L)$ is a matrix polynomial in the lag operator.

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$$\Phi_{FH}(L) = 0 \qquad\qquad B_{FH} = 0$$

- Estimate reduced form for U.S., identify e_t^F.
- Then estimate for U.K.
- Impact of foreign structural shocks on Home variables (B_{HF}): regress $u_t^H = \lambda\, e_t^F$.
  OLS coefficients λ capture the response of H to F structural shocks (Canova, 2005).

Challenge: identifying financial shocks

1. **Fast-moving financial variables** $\Rightarrow$ contemporaneous zero restrictions (Cholesky) implausible; literature moved to sign restrictions.
2. Even with sign restrictions, **financial shocks look like aggregate-demand shocks**: both lead to falling output, prices, and policy rates.
3. **Different financial shocks look identical** to each other, with similar qualitative effects on aggregates. Separating them requires higher VAR dimensionality $\Rightarrow$ *curse of dimensionality*.

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3. **Different financial shocks look identical** to each other, with similar qualitative effects on aggregates. Separating them requires higher VAR dimensionality $\Rightarrow$ *curse of dimensionality*.

$\Rightarrow$ The paper shows it is possible to use a **minimum set of restrictions** to distinguish a broad class of financial shocks from a classical aggregate-demand shock.

Solution: response of the borrowing rate

Based on theoretical models with sticky prices and a financial-accelerator mechanism (Bernanke et al., 1999; Christiano et al., 2014):

$$i_B^m = i^m + x^m$$

- i_B^m: borrowing rate, i^m: government bond rate, x^m: credit spread.

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After an adverse shock, monetary policy eases ($i^m \downarrow$) but the credit spread rises ($x^m \uparrow$), so the response of i_B^m is **theoretically ambiguous**. We exploit this ambiguity:

Negative financial shock: spread dominates $\Rightarrow$ $x^m \uparrow > i^m \downarrow \Rightarrow \boldsymbol{i_B^m} \uparrow$

Negative demand shock: policy rate dominates $\Rightarrow$ $i^m \downarrow > x^m \uparrow \Rightarrow \boldsymbol{i_B^m} \downarrow$

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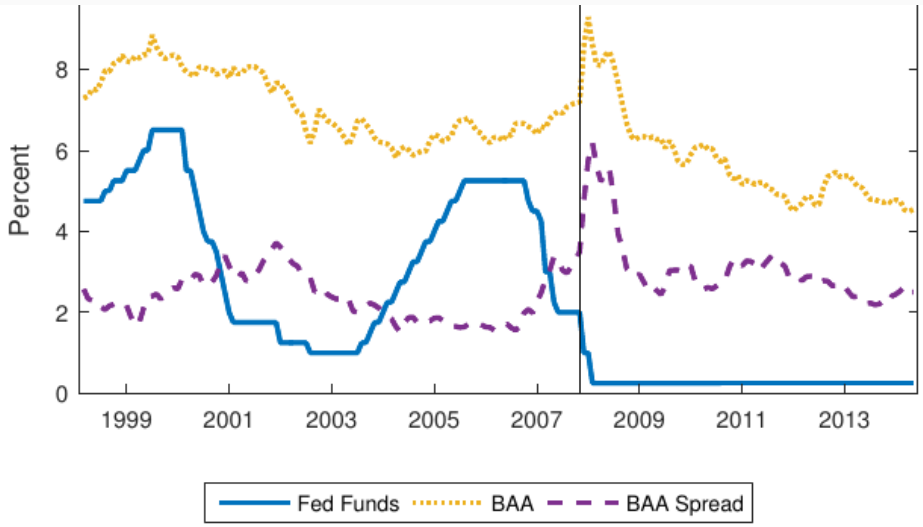
Negative demand shock: policy rate dominates
$\Rightarrow$ $i^m \downarrow > x^m \uparrow \Rightarrow \boldsymbol{i_B^m} \downarrow$ $\Rightarrow$ The **sign of** i_B^m is the minimum restriction that uniquely identifies a broad class of financial shocks from aggregate-demand shocks.

Empirical evidence: the Lehman moment

Figure 1 — Fed Funds, BAA Interest Rate, and BAA Spread



NOTE. The chart reports the Fed Funds rate (solid line), the yield on BAA-rated corporate debt (dotted line), and its spread over a safe interest rate of the same maturity (a 10-year government bond) over the period 1999:M1 to 2015:M3. The vertical line shows 2008:M9, when Lehman Brothers collapsed.

- At the Lehman Brothers collapse (2008:M9): Fed cuts sharply ($i^m \downarrow$) – yet the BAA spread spikes ($x^m \uparrow$).
- **Spread dominates** $\Rightarrow$ the borrowing rate *rises* despite monetary easing.

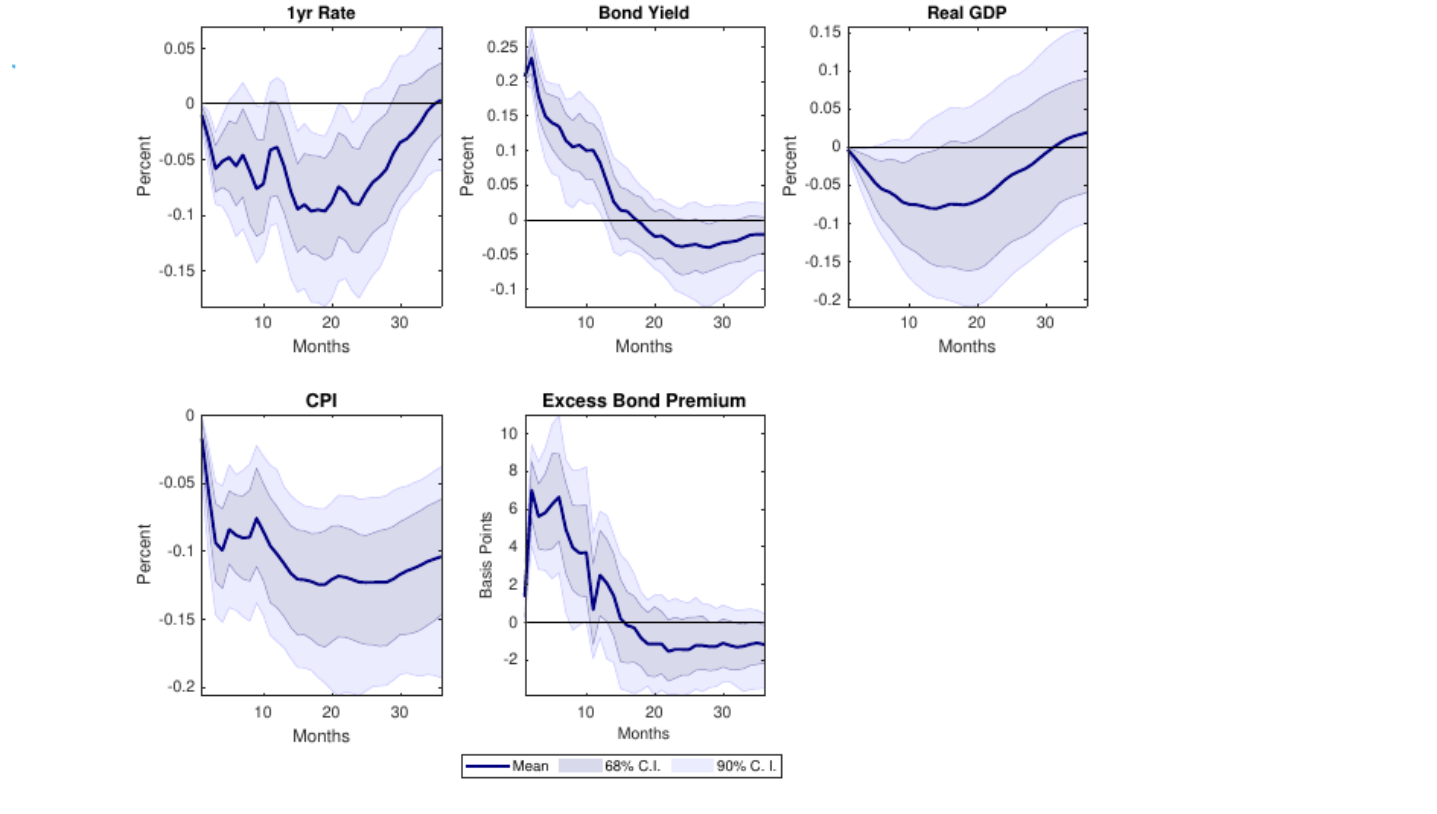
Data & Results

Data and estimation

- **U.S. VAR (Foreign):** real GDP, CPI, 1-yr government bond rate, Excess Bond Premium (Gilchrist & Zakrajšek, 2012), borrowing rate (10-yr gov bond + GZ corporate spread).
- **Identification of MP and CBI shocks:** following Jarocinski & Karadi (2020), decompose high-frequency surprises in Fed-Funds futures (FF4) around FOMC announcements into a monetary-policy and an information component; use as external instruments z_t in the U.S. VAR.
- **U.K. VAR (Home):** real GDP, CPI, 1-yr gilt rate, nominal ER (vs. USD), corporate bond spread; 3 lags of U.S. variables included as exogenous regressors (block exogeneity).
- **Inference:** credible intervals via wild bootstrap (Mertens & Ravn, 2013).

U.S. financial shock: domestic transmission

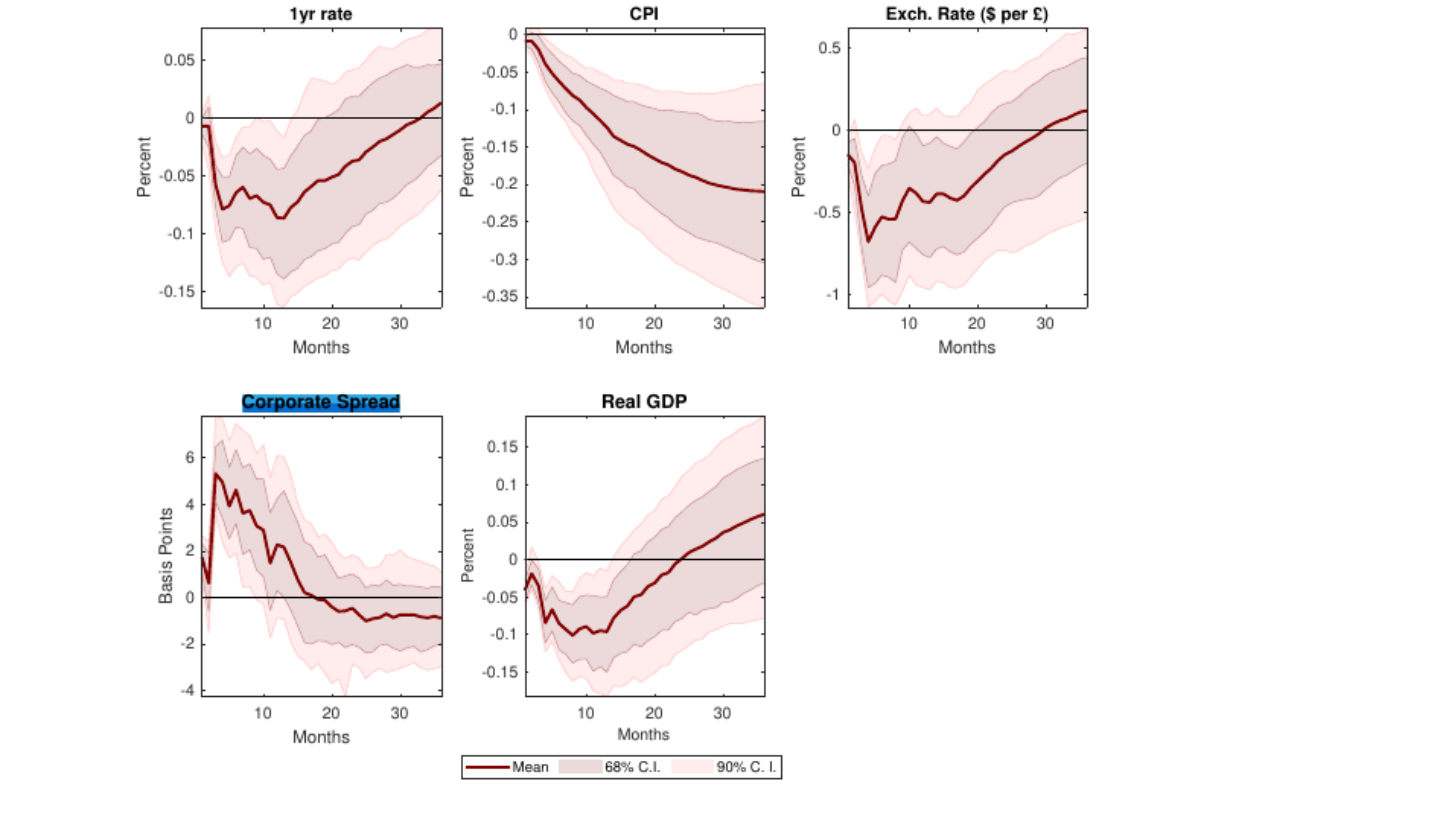
Figure 4 — US Financial Shock: Transmission Within the US



- **1 SD** adverse financial shock: Excess Bond Premium rises (∼7 bp), real GDP contracts (∼0.1%), prices fall (∼0.15%) – consistent with the sign restrictions.
- Despite **loosening** monetary policy (1-yr rate ↓ ∼0.1%), credit conditions **tighten** – the spread dominates, as the identification predicts.

U.S. financial shock: international transmission to the U.K.

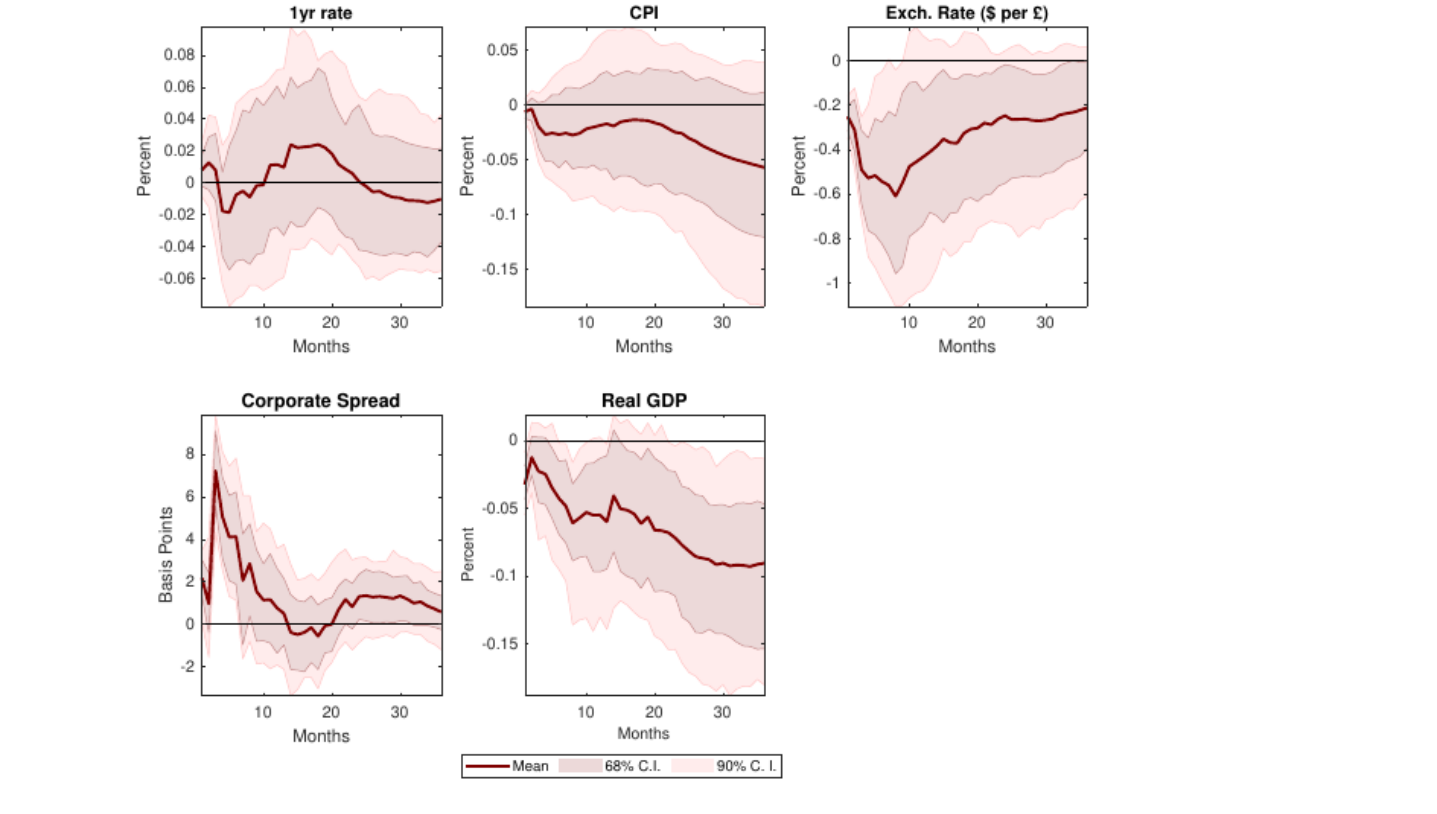
Figure 5 — US Financial Shock: Transmission to the UK



- U.K. corporate spreads rise to ∼5 bp; real GDP and prices fall; sterling depreciates vs. the dollar.
- Happens **despite accommodating U.K. monetary policy** (1-yr gilt rate ↓ ∼10 bp).
- ⇒ Evidence of an **international credit channel** of financial shocks.

U.S. monetary-policy shock: transmission to the U.K.

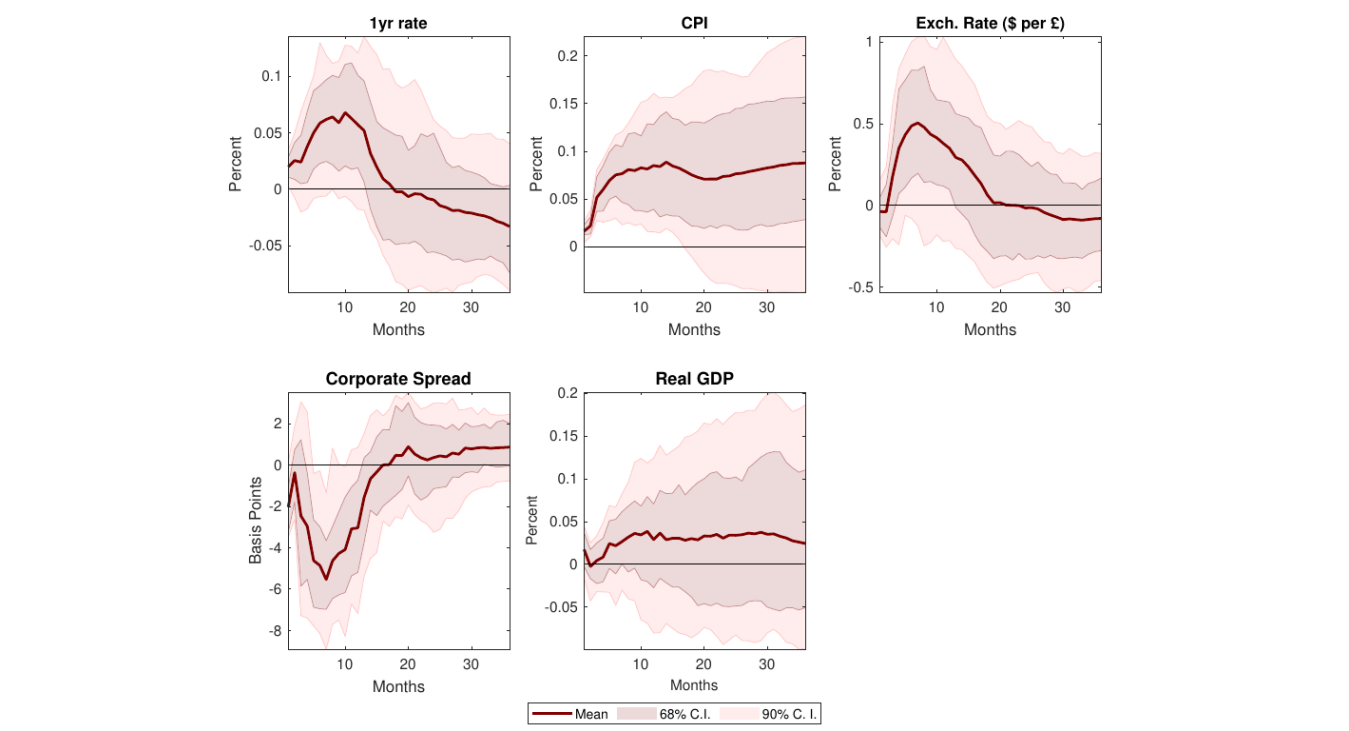
Figure 7 — US Monetary Policy Shock: Transmission to the UK



- U.S. MP tightening raises U.K. corporate spreads (∼7 bp); U.K. real GDP contracts; sterling depreciates.
- Consistent with the literature (Rey, 2016; Kalemli-Özcan, 2019), with an improved identification that **explicitly controls for CBI effects**.

Central-bank information shock: transmission to the U.K.

Figure 9 — Fed Information Shock: Transmission to the UK



- A positive CBI shock (better-than-expected U.S. outlook) **compresses** U.S. credit spreads $\Rightarrow$ U.K. spreads fall, real GDP and prices rise.
- **Symmetry** of the ICC: the same mechanism that amplifies a bad shock also amplifies a good shock.

(Results hold when identifying an additional supply shock, using the inflation-targeting subsample from 1993:M1, excluding the Global Financial Crisis, and replacing GDP with unemployment.)

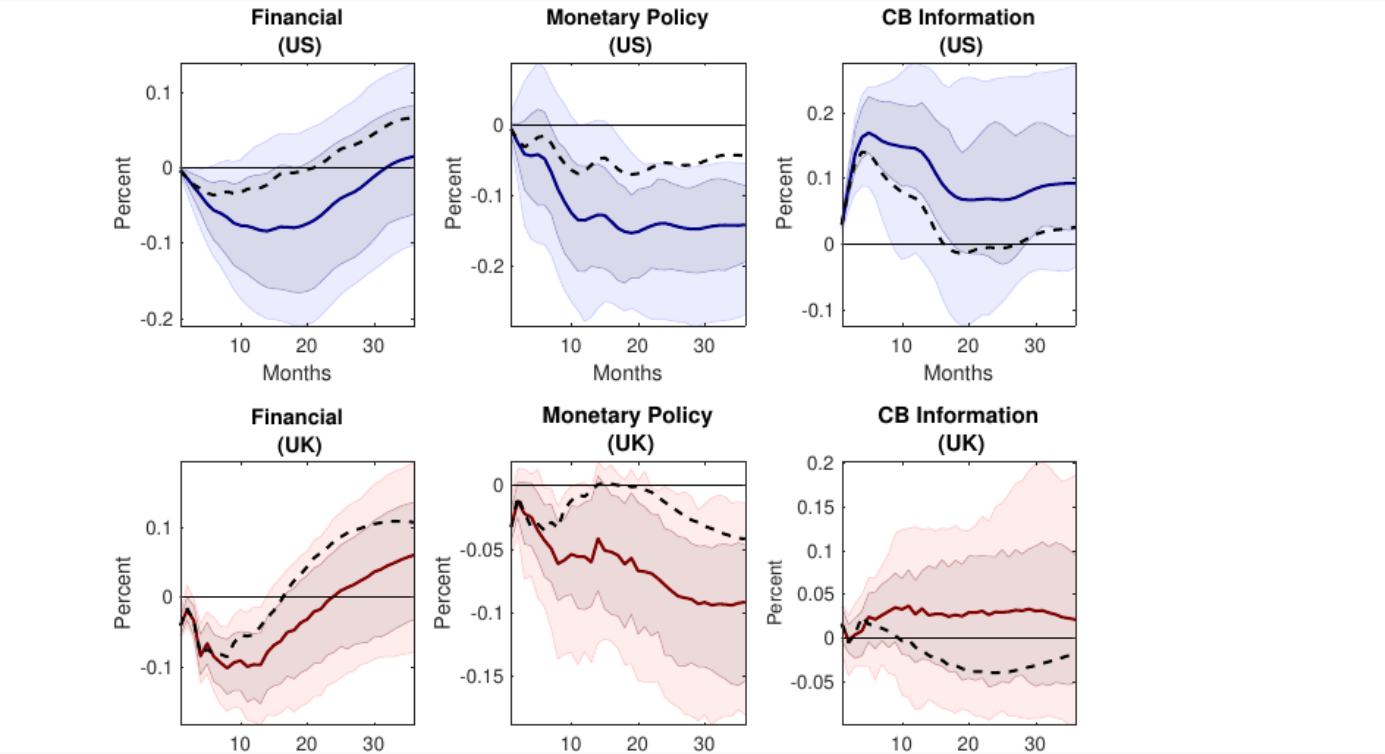
Mechanism

How do credit spreads amplify international transmission?

- When an international credit channel is at work, domestic and foreign credit spreads not only **co-move** but also **amplify the real effects** of foreign shocks.
- **The mechanism – a feedback loop:**
  - Adverse shock $\rightarrow$ credit spreads rise,
  - higher borrowing costs depress demand,
  - lower demand reduces net worth,
  - lower net worth raises spreads further.
- With **internationally integrated balance sheets**, this loop operates across borders – no-arbitrage in integrated financial markets creates tight linkages in macro dynamics across countries.

Inspecting the mechanism

Figure 10 — Closing the International Credit Channel: The Response of Real GDP



- **Counterfactual holding credit spreads fixed:** GDP responses are *smaller and less persistent* for all three shocks – confirming the amplification role of credit spreads.
- **Alternative spread measures** (mortgage, Libor) respond counter-cyclically to all three shocks – the ICC is not specific to corporate bond spreads.

Discussion

- The **ICC is a general transmission channel**, operating through at least three distinct U.S. shocks:
  - financial shocks, monetary-policy shocks, and central-bank information shocks.
- The channel is **symmetric** – same mechanism amplifies both contractionary and expansionary spillovers.
- **Implications:**
  - Domestic monetary policy alone cannot insulate open economies from foreign financial conditions (cf. Rey's trilemma $\rightarrow$ dilemma).
  - Credit-market stabilisation is first-order for open-economy macro policy.

Conclusion

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- **New identification** combining high-frequency IVs with a minimal sign restriction on the *borrowing rate* – cleanly separates financial shocks from aggregate-demand shocks.
- **International Credit Channel is pervasive:** U.S. financial, monetary-policy, and CBI shocks all transmit to the U.K. via credit spreads.
- **Credit spreads amplify** the international transmission of U.S. shocks – closing them in a counterfactual materially dampens U.K. responses.
- **Policy takeaway:** in financially integrated economies, foreign credit conditions are a first-order driver of domestic macro dynamics.

Thank you!

Comments and feedback welcome.

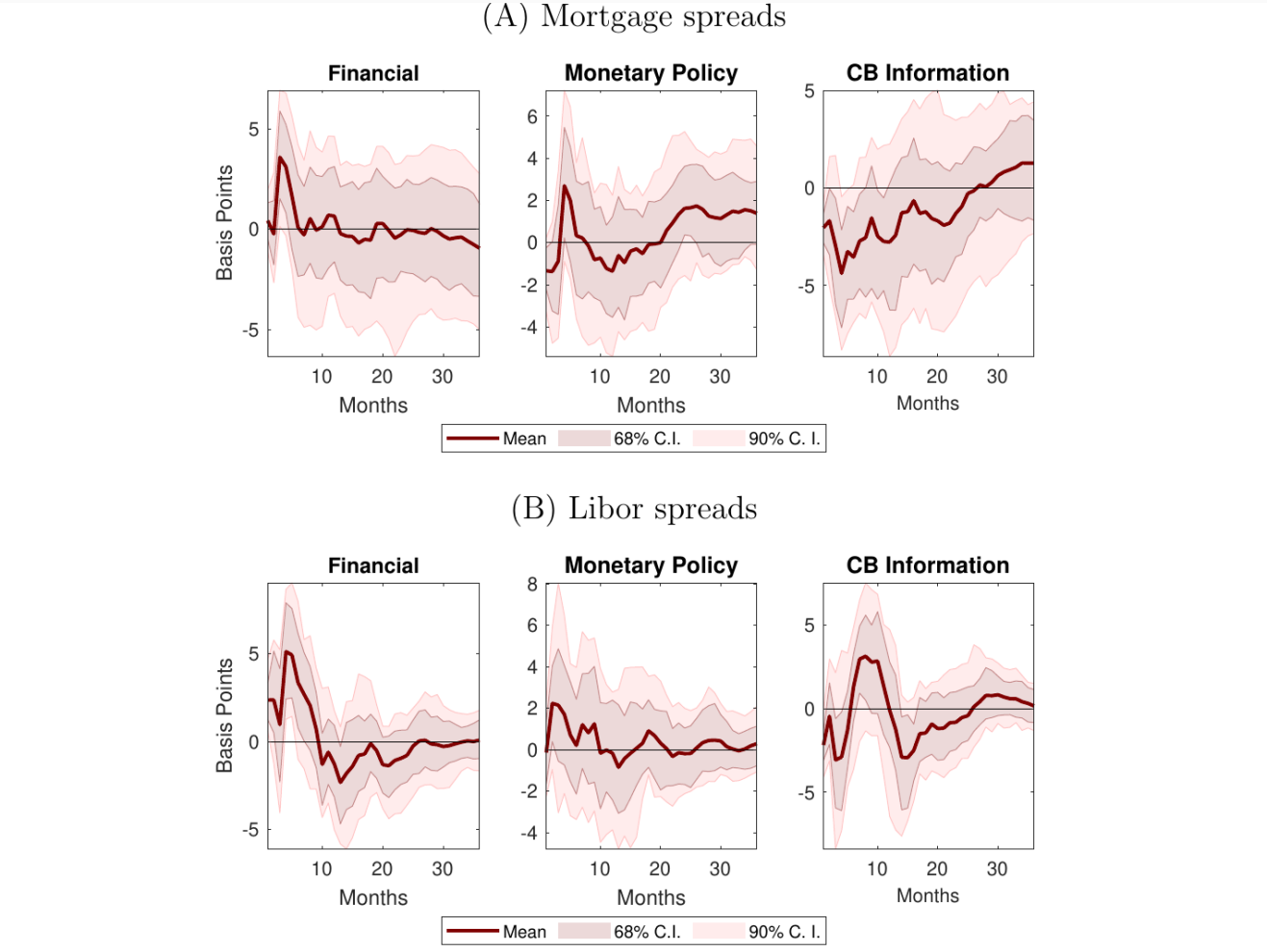
Appendix: robustness

- Identify an additional supply shock.
- Restrict to the inflation-targeting subsample (from 1993:M1).
- Exclude the Global Financial Crisis window.
- Replace real GDP with unemployment.

Main results unchanged across all alternatives.

Appendix: alternative credit-spread measures

Figure 11 — UK's Response to US Financial Shock: Alternative Credit Spread Measures



Top panel: mortgage spreads. Bottom panel: Libor spreads. Both respond counter-cyclically